

利用paml批量计算基因的dn ds
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理论原理

Chapter

1

theory

dn ds 简介

1. dn/ds 仅对编码区而言，测量的是一段编码区域而不是某一个位点。
2. 同义突变的积累速率， ds 被认为是中性的，仅仅和突变率相关。
3. 非同义突变的积累速率， dn 被认为和突变率，以及群体大小相关。
4. dn/ds 被认为和选择与群体大小相关。
 1. $dn/ds > 1$ 正选择
 2. $dn/ds < 1$ 负选择
 3. $dn/ds = 1$ 中性进化

计算方法

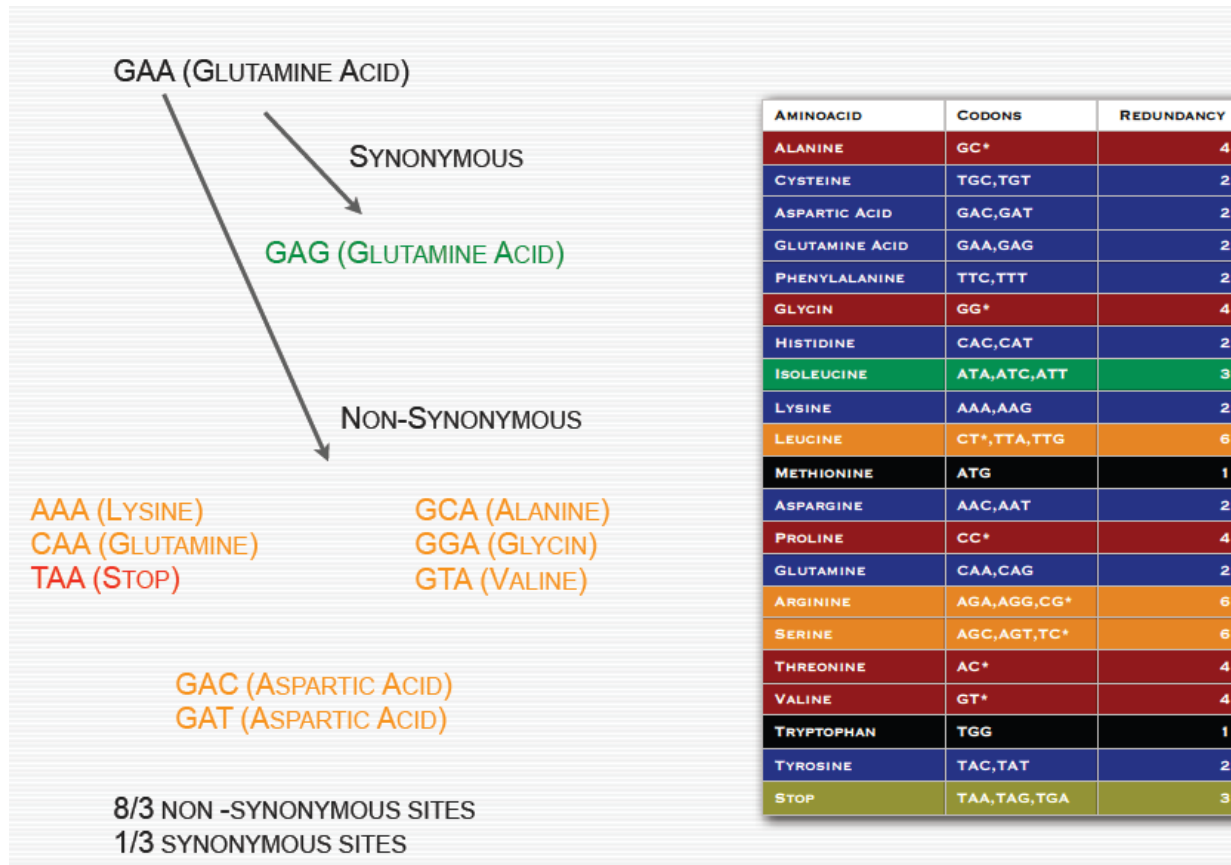
1. dn ds大致的计算思路:

ds≈被固定的同义突变的数量/随机突变是同义的比例

dn≈被固定的非同义突变数量/随机突变时非同义的比例

例

比如:



ACA	ATA	ATC	TTT	AAT	CAA
T	I	I	F	N	Q
ACA	ATA	ACC	TTT	AAC	CAA
T	I	T	F	N	Q

Observed 1 synonymous and 1 non-synonymous substitution (observed S/N ratio of 1). What is expected under neutrality?

Seq1	ACA	ATA	ATC	TTT	AAT	CAA
Syn	1	2/3	2/3	1/3	1/3	1/3
NonSyn	2	7/3	7/3	8/3	8/3	7/3
Seq2	ACA	ATA	ACC	TTT	AAC	CAA
Syn	1	2/3	1	1/3	1/3	1/3
NonSyn	2	7/3	2	8/3	8/3	7/3
Mean						
Syn	1	2/3	5/6	1/3	1/3	1/3
NonSyn	2	7/3	13/6	8/3	8/3	7/3

ES = 3 1/2, EN = 14 1/6: under neutrality, would expect the ratio of non-synonymous to synonymous substitutions of EN/ES ~ 4

根据nei-gojobori模型，N:S/EN:ES=dn/ds=0.25,所以该编码序列受到纯化选择。

- 一般计算kaks是两个或多个物种比出一个kaks值， branch-site 是对进化树上多个物种分别和他们的祖先物种比较计算出各个分支的kaks。一般先输入一个树结构， paml不会用支长但会用结构。

分析方法

Chapter 2

Practical experiment

CalculateKaKsusingMusclePaml.py的程序逻辑

- 循环计算每一组同源基因的ka/ks（每一组可以有多个同源基因）
 - 首先调用muscle对同源基因的氨基酸序列进行多序列比对
 - 根据cds.fa序列回填出cds的多序列伪比对（即根据氨基酸的位置的比对）
 - 调用paml计算该基因的 dn ds dn/ds
- 以下情况的基因被忽略（条件可以调整）：
 - 同源基因当中某基因氨基酸序列的长度小于 -l 指定的长度
 - 同源基因当中氨基酸序列有超过2倍的长度差异
 - 同源基因当中有终止子出现在序列当中
- 目前读取paml输出结果采用DIY的方式。
- 两个程序同时运行，不要在同一个目录下跑

运行命令及说明

- `python /opt/life/src/bioinfodevelop/analysisAppEntry/CalculateKaKsusingPaml.py -c /opt/pubdata/kakawork24primate/kaks.config -m /opt/pubdata/kakawork24primate/primate7.orthologous.ChA.BmG.Oar.NLeu.GSM.step1out.pephomo -p /opt/pubdata/kakawork24primate/proteinCDSpair.txt -l 10`
- Config文件指定：
 - 调用的相关软件路径。
 - `processType`指定以A/B/C那种模式运行（branch-site/branch/pairwise comparison model）
 - 存在`pamltreefile`参数，则paml每次计算都用`pamltreefile`中的树文件。注释掉该参数，则paml每次计算都自己根据序列建树。
 - 如果指定了`fgspecies`,`outgroup`(必须同时存在)，则`pamltreefile`不存在或`pamltreefile`中没有`#1/$1`指定前景支。此时以`outgroup`为`backgroundbranch`对每个`fgspecies`计算
 - 如果没有指定`fgspecies`,`outgroup`，则根据`pamltreefile`中是否设定了`forebranch`，决定按照`pamltreefile`设定运行每个基因的计算或者每个基因都依次选取各个物种作为`forebranch`计算(比如`pamltreefile`不存在或没有设定`forebranch`)

```
st ~]$ python /opt/life/src/bioinfodevelop/analysisAppEntry/CalculateKaKsusingPaml.py -h
loading config /opt/life/src/config.py 192.168.3.6
Usage: CalculateKaKsusingPaml.py [options]

Options:
  -h, --help                show this help message and exit
  -m HOMOLOGOUSGENE, --homologousgene=HOMOLOGOUSGENE
                           homologous file
  -p PROTEINCDSPAIRFILE, --proteincdspair=PROTEINCDSPAIRFILE
                           proteincdspairfile
```

- 配置文件:

```
[software]
musclepath=/opt/bioinfoTools/muscle3.8.31_i86linux64
pamlpath=/opt/bioinfoTools/paml4.9j
PhyMLpath=/opt/bioinfoTools/PhyML
clustalw=/opt/bioinfoTools/clustalw-2.1-static/clustalw2
[parameter]
pamltreefile=/opt/pubdata/kakawork24primate/primate7.specieslist.nwk_fixmarkbran
ch
ctltemplatefile=/opt/bioinfoTools/paml4.9j/codeml.ctl.template.ctl
outfilepre=/opt/pubdata/kakawork24primate/primate7
processType=B
#fgspecies=,,,
#outgroup=
~
```

为空则在当前运行目录下输出

- 蛋白序列和对应CDS序列文件:

```
BmG:/opt/pubdata/24pep/Bos_mutus.BosGru_v2.0.pep.all.modifiedtitle.fa.corrected;/opt/pubdata/24cds/Bos_mutus.BosGru_v2.0.cds.all.modifiedtitle.fa
ChA:/opt/pubdata/24pep/Capra_hircus.ARS1.pep.all.modifiedtitle.fa.corrected;/opt/pubdata/24cds/Capra_hircus.ARS1.cds.all.modifiedtitle.fa
GSM:/opt/pubdata/24pep/Rhinopithecus_roxellana.Rrox_v1.pep.all.modifiedtitle.fa.corrected;/opt/pubdata/24cds/Rhinopithecus_roxellana.Rrox_v1.cds.all.modifiedtitle.fa
HMol:/opt/pubdata/24pep/Hylobates_moloch_pep.modifiedtitle.fa;/opt/pubdata/24cds/Hylobates_moloch_cds_from_genomic.modifiedtitle.fa
lcJW:/opt/pubdata/24pep/Lemur_catta.modifiedtitle.fa;/opt/pubdata/24cds/GCF_020740605.2_mLemCat1.pri_cds_from_genomic.modifiedtitle.fa
MSP:/opt/pubdata/24pep/Masph.final.gene.pep.modifiedtitle.fa;/opt/pubdata/24cds/Masph.final.gene.cds.modifiedtitle.fa
NLeu:/opt/pubdata/24pep/Nomascus_leucogenys.Nleu_3.0.pep.all.modifiedtitle.fa.corrected;/opt/pubdata/24cds/Nomascus_leucogenys.Nleu_3.0.cds.all.modifiedtitle.fa
Oar:/opt/pubdata/24pep/Ovis_aries_rambouillet.Oar_rambouillet_v1.0.pep.all.modifiedtitle.fa.corrected;/opt/pubdata/24cds/Ovis_aries_rambouillet.Oar_rambouillet_v1.0.cd
.all.modifiedtitle.fa
Ovir:/opt/pubdata/24pep/GCF_002102435.1_Ovir.te_1.0_protein.modifiedtitle.fa;/opt/pubdata/24cds/GCF_002102435.1_Ovir.te_1.0_cds_from_genomic.modifiedtitle.fa
sCA:/opt/pubdata/24pep/Sapajus_apella_pep.modifiedtitle.fa;/opt/pubdata/24cds/Sapajus_apella_cds_from_genomic.modifiedtitle.fa
Thy:/opt/pubdata/24pep/Trachypithecus_francoisi_pep.modifiedtitle.fa;/opt/pubdata/24cds/Trachypithecus_francoisi_cds_from_genomic.modifiedtitle.fa
```

- 蛋白序列fa文件和cds序列fa文件都需要修改title,是transcript:xxxx可以相互对应。并且同源基因列表也要基于此transcript ID号制作。
- 格式为:
物种名: 修改后的氨基酸fa文件全路径; cds fa文件的全路径

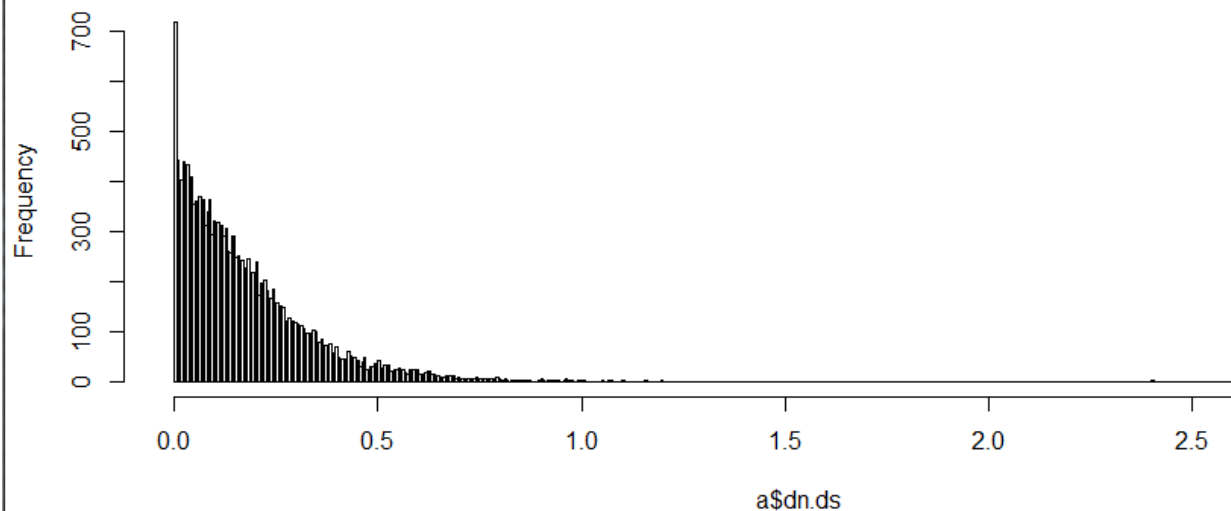
结果收集展示

- 每组同源基因，几种fg设置方式，就有几个fg/bg dnds 对，就有几个ds文件
 - tpid fg1dnds bg1dnds fg2dnds bg2dnds
- 对fg1/bg1 的ds文件
 - Tpid sp1ds sp2ds sp3ds ...

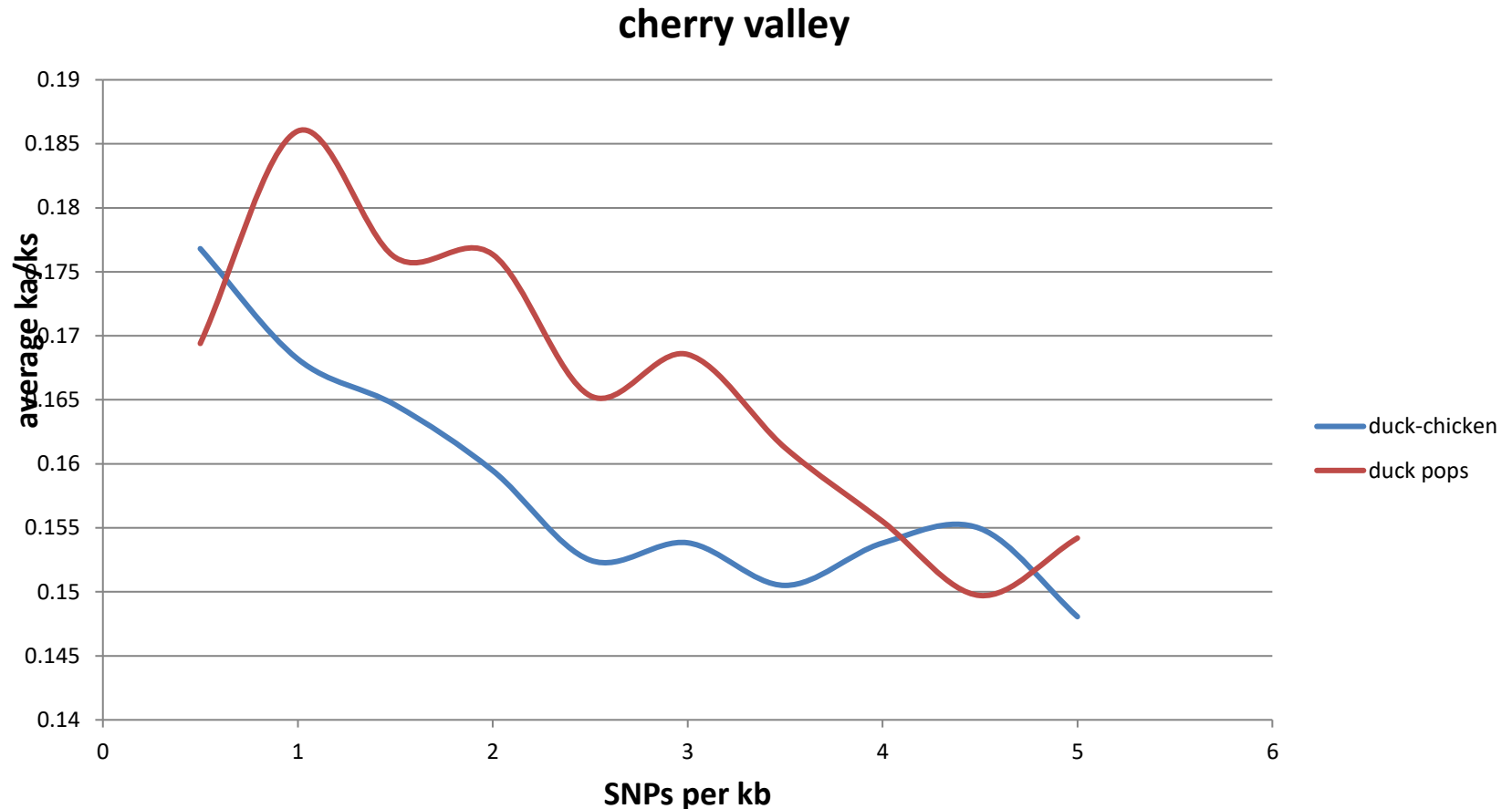
2014experiment输出结果示例

- 人与猪的同源基因计算dn ds

pig	human	dn/ds	dn	ds
ENSSSCT00000033414	ENST00000390534	0.0083	0.5051	60.9578
ENSSSCT00000033467	ENST00000390242	0.3907	0.0723	0.1851
ENSSSCT00000010999	ENST00000390321	0.1602	0.1497	0.9343
ENSSSCT00000008984	ENST00000390237	0.3124	0.2640	0.8451
ENSSSCT00000033546	ENST00000390305	0.2692	0.1952	0.7252
ENSSSCT00000034850	ENST00000390305	0.2232	0.1190	0.5332
ENSSSCT00000036351	ENST00000390319	0.3693	0.1795	0.4861
ENSSSCT00000036101	ENST00000390283	0.3984	0.2053	0.5152
ENSSSCT00000008985	ENST00000417279	0.5896	0.3686	0.6251
ENSSSCT00000027499	ENST00000453166	0.3363	0.1292	0.3843
ENSSSCT00000032222	ENST00000491977	0.3363	0.1544	0.4590
ENSSSCT00000033135	ENST00000390283	0.3692	0.1742	0.4717
ENSSSCT00000033876	ENST00000390283	0.2991	0.1461	0.4884
ENSSSCT00000030345	ENST00000468879	0.4979	0.1734	0.3482



Genes in SNP deserts have higher KA/KS ration



Compute ka/ks between chicken and duck 1:1 homo

gene: 15066 genes take in to count

Compute ka/ks within 7 duck populations: 15066

genes take in to count

数据准备

Chapter

3

preprocess

蛋白序列和对应CDS序列文件title修改

- 从ensembl上下载要比较物种的cds与氨基酸序列fa文件
- 将氨基酸序列fa文件的title改成
“>gene:ENSG00000237235|transcript:ENST00000434970”形式。
 - 命令为: `awk '/gene:/{printf ">$4"|" $5"\n";next}$0!~/gene:/{printf $0"\n";next}' Homo_sapiens.GRCh38.pep.all.fa>Homo_sapiens.GRCh38.pep.all.fa_modify`
 - (对CatalogSNP_withaaseq_cdsseq.py生成的aa文件: `awk '{if($1~/>/){a=substr($0,2);printf ">transcript:%s\n",a}else{print $0}}'`)
- 将cds.fa序列title改成
- “>ENSAPLT00000001579 transcript:ENSAPLT00000001579”
 - `less -S ~/update/new/pep/Nomascus_leucogenys.Nleu_3.0.cds.all.fa.gz|awk '/gene:/{sub(/>/,"transcript:",$1);printf ">$4"|" $1"\n";next}$0!~/gene:/{printf $0"\n";next}' -> Nomascus_leucogenys.Nleu_3.0.cds.all.modified.fa`
- 制作同源基因列表: (两个物种的简易方式)
- `makeblastdb -in input_file -dbtype molecule_type -title database_title -parse_seqids -out database_name -logfile`
 - 建库 `makeblastdb -in Sus_scrofa.Sscrofa10.2.75.pep.all -dbtype prot -parse_seqids -hash_index -out Sus_scrofa.Sscrofa10.2.75.pep.all`
 - `blastp -query Sus_scrofa.Sscrofa10.2.75.pep.all.modified.fa -task blastp -db Sus_scrofa.Sscrofa10.2.75.pep.all -out human_pig_pep_compare -outfmt 7 -max_target_seqs 1 -num_threads 20`

- ~~此处title~~需要手工加上，与下面的物种名一致

从新版本开始，兼容如下格式，支持：

同源基因表中，同源组有物种无同源基因（用NA 或 None表示）

同源组中，各个物种存在多个同源基因（用,或;分隔的基因ID）

File	GSN	Hmol	lclW	MSF	NLEuc	sCA
XP_033032570.1	XP_033032971.1	XP_033033494.1	XP_033057432.1	XP_033057433.1	XP_033057434.1	XP_033071161.1
XP_033033662.1	XP_033033663.1	XP_033059179.1	XP_033072346.1	XP_033072347.1	XP_033081848.1	XP_033081850.1
XP_03307637.1	XP_03307638.1	XP_03307639.1	XP_03307640.1	XP_03307641.1	XP_03307642.1	XP_03307643.1
XP_033034146.1	XP_033034147.1	XP_033034148.1	XP_033034149.1	XP_033034150.1	XP_033034151.1	XP_033034152.1
XP_033010315.1	XP_033010316.1	XP_033010317.1	XP_033010318.1	XP_033010319.1	XP_033010320.1	XP_033010321.1
XP_033056277.1	XP_033053979.1	XP_033053980.1	XP_033053981.1	XP_033053982.1	XP_033053983.1	XP_033053984.1
XP_033080457.1	XP_033080458.1	XP_033080459.1	XP_033080460.1	XP_033080461.1	XP_033080462.1	XP_033080463.1
XP_03305908.1	XP_03305909.1	XP_03305910.1	XP_03305911.1	XP_03305912.1	XP_03305913.1	XP_03305914.1
XP_033071840.1	XP_033085112.1	XP_033085113.1	XP_033085114.1	XP_033085115.1	XP_033085116.1	XP_033085117.1
XP_033034899.1	XP_033034910.1	XP_033034911.1	XP_033034912.1	XP_033034913.1	XP_033034914.1	XP_033034915.1
XP_033067737.1	XP_033067738.1	XP_033067739.1	XP_033067740.1	XP_033067741.1	XP_033067742.1	XP_033067743.1
XP_033041665.1	XP_033041666.1	XP_033041667.1	XP_033041668.1	XP_033041669.1	XP_033041670.1	XP_033041671.1
XP_0330363584.1	XP_033036358.1	XP_033036359.1	XP_033036360.1	XP_033036361.1	XP_033036362.1	XP_033036363.1
XP_033075374.1	XP_033075375.1	XP_033075376.1	XP_033075377.1	XP_033075378.1	XP_033075379.1	XP_033075380.1
XP_033065763.1	XP_033065764.1	XP_033065765.1	XP_033065766.1	XP_033065767.1	XP_033065768.1	XP_033065769.1
XP_033064814.1	XP_033064824.1	XP_033064833.1	XP_033064843.1	XP_033064853.1	XP_033064863.1	XP_033064873.1
XP_033094103.1	XP_033094104.1	XP_033094105.1	XP_033094106.1	XP_033094107.1	XP_033094108.1	XP_033094109.1
XP_033075719.1	XP_033075720.1	XP_033075721.1	XP_033075722.1	XP_033075723.1	XP_033075724.1	XP_033075725.1
XP_033054636.1	XP_033054637.1	XP_033054638.1	XP_033054639.1	XP_033054640.1	XP_033054641.1	XP_033054642.1
XP_033064000.1	XP_033064001.1	XP_033064002.1	XP_033064003.1	XP_033064004.1	XP_033064005.1	XP_033064006.1
XP_033058216.1	XP_033058217.1	XP_033058218.1	XP_033058219.1	XP_033058220.1	XP_033058221.1	XP_033058222.1
XP_033064506.1	XP_033064507.1	XP_033064508.1	XP_033064509.1	XP_033064510.1	XP_033064511.1	XP_033064512.1
XP_033075531.1	XP_033075532.1	XP_033075533.1	XP_033075534.1	XP_033075535.1	XP_033075536.1	XP_033075537.1
XP_033011972.1	XP_033011973.1	XP_033011974.1	XP_033011975.1	XP_033011976.1	XP_033011977.1	XP_033011978.1
XP_033081324.1	XP_033081325.1	XP_033081326.1	XP_033081327.1	XP_033081328.1	XP_033081329.1	XP_033081330.1
XP_033014543.1	XP_033014544.1	XP_033014545.1	XP_033014546.1	XP_033014547.1	XP_033014548.1	XP_033014549.1
XP_033060597.1	XP_033060598.1	XP_033060599.1	XP_033060600.1	XP_033060601.1	XP_033060602.1	XP_033060603.1
XP_033034316.1	XP_033034317.1	XP_033034318.1	XP_033034319.1	XP_033034320.1	XP_033034321.1	XP_033034322.1
XP_033073721.1	XP_033073722.1	XP_033073723.1	XP_033073724.1	XP_033073725.1	XP_033073726.1	XP_033073727.1
XP_033041397.1	XP_033041398.1	XP_033041399.1	XP_033041400.1	XP_033041401.1	XP_033041402.1	XP_033041403.1
XP_033050443.1	XP_033050444.1	XP_033050445.1	XP_033050446.1	XP_0		

但仅计算forebranch基因全部存在无缺失，backgroundbranch不全缺失的同源组。且每个物种如存在多个同源基因，仅取其中一个

Acromyrmex echinaior(103372)			Acyrtosiphon pisum(7029)			Aedes aegypti(7159)			Aegilops tauschii(37682)			Agaricus bisporus(5341)			Alluropoda melanoleuca(9646)			Albugo laibachii(653948)			Alligator mississippiensis(84)		
XP_033030880.1	*F4WHB5	*ACYPI001768-PA	*AAEL017515-PA	NA	NA	*ENSAMEP00000008342	NA	NA	*XP_006259526	*XP_006036638	NA	NA	NA	NA	*ENSACAP00000000957	*ADAC001191-PA	*AGAP000237-PA	NA	NA	*GB46492-PA	NA	NA	
XP_033030881.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030882.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030883.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030884.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030885.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030886.1	NA	NA	NA	NA	NA	*ENSAMEP00000006858	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
XP_033030887.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030888.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030889.1	*F4W8A5	*F4W9Q0	*ACYPI002888-PA	*ACYPI008980-PA	*AAEL006710-PA	*AAEL009072-PA	NA	NA	*ENSAMEP00000016968	NA	NA	*XP_006272065	*XP_006032964	NA	NA	NA	NA	NA	NA	*ADAC000209-PA	*ADAC005481-PA	NA	
XP_033030890.1	*F4WJ56	*ACYPI001766-PA	NA	NA	NA	*ENSAMEP000000016911	NA	NA	*XP_006268797	*XP_006032962	NA	NA	*PAC_15728314	*ENSACAP000000007753	NA	NA	NA	NA	NA	*GB49350-PA	*XP_005110758	NA	
XP_033030891.1	*F4X4B6	*ACYPI008114-PA	*AAEL008719-PA	NA	NA	NA	NA	NA	*XP_006032961	NA	NA	*PAC_15725370	*ENSACAP000000007711	*ADAC003166-PA	*AGAP002706-PA	NA	NA	NA	NA	*GB50044-PA	NA	NA	
XP_033030892.1	*F4X4B6	*ACYPI008114-PA	*AAEL008719-PA	NA	NA	NA	NA	NA	*XP_006032961	NA	NA	*PAC_15725370	*ENSACAP000000007711	*ADAC003166-PA	*AGAP002706-PA	NA	NA	NA	NA	*GB50044-PA	NA	NA	
XP_033030893.1	*F4X4B6	*ACYPI008114-PA	*AAEL008719-PA	NA	NA	NA	NA	NA	*XP_006032961	NA	NA	*PAC_15725370	*ENSACAP000000007711	*ADAC003166-PA	*AGAP002706-PA	NA	NA	NA	NA	*GB50044-PA	NA	NA	
XP_033030894.1	*F4X4B6	*ACYPI008114-PA	*AAEL008719-PA	NA	NA	NA	NA	NA	*XP_006032961	NA	NA	*PAC_15725370	*ENSACAP000000007711	*ADAC003166-PA	*AGAP002706-PA	NA	NA	NA	NA	*GB50044-PA	NA	NA	
XP_033030895.1	NA	NA	NA	NA	NA	*ENSAMEP00000006858	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
XP_033030896.1	*F4X3K1	*ACYPI004892-PA	*AAEL007414-PA	*AAEL015446-PA	NA	NA	*ENSAMEP000000016885	NA	NA	*XP_006262635	*XP_006032960	NA	NA	*PAC_15719648	NA	*ADAC004750-PA	*AGAP005979-PA	NA	NA	*GB52	NA	NA	
XP_033030897.1	NA	NA	NA	NA	NA	*ENSAMEP000000016877	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	

out.emapper.orthologs.simplify.reform

程序根据同源基因表中的（物种对应的）基因ID，在（各物种）pep和cds序列文件中寻找序列并计算kaks。如果pep和cds中的ID是在特定提示字符串之后，比如‘transcript:’之后的id为表中对应的id，最好通过-s transcript: 指定。否则程序根据同源表中id在pep和cds序列文件中找到>title中包含id的序列。

processType=B/A需要指定pamltreefile,通过#1标记froebbranch。其他指定方式暂未支持。

鉴定多物种同源基因(多种方法流程)

- **Orthomcl(2003)**
- Orthofinder()
- OMA standalone
- Ensembl pipeline
- Mestortho(2008)

OrthoMCL(已安装mariadb/mysql)

- 1) 下载14各物种的pep序列（使用修改title之后的）
- 2) 创建名字不同的新的数据库，以防之前数据被覆盖
- 3) `orthomclInstallSchema orthomcl.config.template`

```
# this config assumes a mysql database named 'orthomcl'.  adjust according
# to your situation.
dbVendor=mysql
dbConnectionString=dbi:mysql:orthomcl 新数据库名
dbLogin=root
dbPassword=
similarSequencesTable=SimilarSequences
orthologTable=Ortholog
inParalogTable=InParalog
coOrthologTable=CoOrtholog
interTaxonMatchView=InterTaxonMatch
percentMatchCutoff=50
evaluateExponentCutoff=-5
oracleIndexTblSpc=NONE
```

4) 修改为orthomcl输入格式

```
less -S Bos_mutus.BosGru_v2.0.pep.all.modifiedtitle.fa | orthomclAdjustFasta BGu - 2
```

```
less -S
```

```
Ovis_aries_rambouillet.Oar_rambouillet_v1.0.pep.all.modifiedtitle.fa | orthomclAdjustFasta  
Oar - 2
```

◦ ◦ ◦

```
mv *fasta orthomclAdjustFast.out #整理到新建的orthomclAdjustFast.out 文件夹
```

5) 过滤数据(长度等质控)

得到文件

goodProteins.fasta

poorProteins.fasta

```
[lrui@localhost 24pep]$ orthomclFilterFasta orthomclAdjustFast.out 10 20  
processing file BmG.fasta  
processing file ChA.fasta  
processing file GSM.fasta  
processing file HMol.fasta  
processing file lcJW.fasta  
processing file MSP.fasta  
processing file NLeu.fasta  
processing file Oar.fasta  
processing file Ovir.fasta  
processing file sCA.fasta  
processing file TFhy.fasta
```

6) Blast (3天+)

- 建库

- all-to-all 比对

```
makeblastdb -title primate24nurt -in goodProteins.fasta -dbtype prot -parse_seqids -out  
orthomclprimate24nurt -logfile primate24.log
```

```
nohup /opt/bioinfoTools/ncbi-blast-2.13.0+/bin/blastp -query goodProteins.fasta -db
```

```
orthomclprimate24nurt -seg yes -out orthomclprimate24 blastout -evalue 1e-5 -outfmt 7
```

7) 处理比对结果数据格式

```
nohup orthomclBlastParser orthomcl14.format.blastout compliantFasta14 >  
orthomcl14.similarSequences &
```

```
perl -p -i -e 's/0\t0/1\t-181/' orthomcl14.similarSequences
```

8) 将比对结果导入数据库

```
orthomclLoadBlast ../orthomcl.config14.template orthomcl14.similarSequences
```

9) 寻找相互最佳比对的成对蛋白（算法、阈值）5天

```
orthomclPairs ../orthomcl.config14.template orthomcl14_pairs.log cleanup=no &
```

10) 从数据库导出

11) 对pairs聚类类，对聚类结果编号。

```
mcl mclInput --abc -l 1.5 -o mclOutput
```

```
orthomclMclToGroups led 1 < mclOutput14 > groups14.txt
```

24个pramite物种项目同源基因列表文件制作

- 从orthomcl的同源结果文件提出1-to-1的结果:

```
python ~/life/src/NGS/SwissArmyKnife/one2oneOrtholog.py -i  
primate7maml4.orthologous -s 1 -t TFhy -t GSM -b BmG -b ChA -  
b HMol -b lcJW -b MSP -b NLeu -b Oar -b Ovir -b sCA
```

```
python ~/life/src/NGS/SwissArmyKnife/one2oneOrtholog.py -i  
primate7maml4.orthologous -s 1 -t TFhy -t GSM -b HMol -b lcJW  
-b MSP -b NLeu -b sCA(每个物种都只有1拷贝)
```

- 制作-m的输入:

```
sed 's/^orthologous.*:\s/gene:/'  
primate7maml4.orthologous.step1out | sed 's/\s/\t~\tgene:/g'  
| less -S > primate7maml4.orthologous.step1out.pephomo
```

关于one-to-one orthologous gene

- <https://www.biostars.org/p/50979/>
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3228760/>
- 结合审的那篇稿子中的分析，6786-4091，
1686个single copy gene

同时修改同源基因表和pep.modifiedtitle.fa 的transcript 号

- 发现修改pep的title命令写错了（OrthoMCL的第一步用了cds的命令），transcript:用了pep的id而应该用原>中transcript:的id。
- 依次修改各个物种相应文件的transcript 号。
 1. `python ~/basicutil/src/bruteForceSubTranscriptID.py -r Capra_hircus.ARS1.pep.all.fa ChA -m primate7maml4.orthologous -t Capra_hircus.ARS1.pep.all.modifiedtitle.fa`
 2. `python ~/basicutil/src/bruteForceSubTranscriptID.py -r Bos_mutus.BosGru_v2.0.pep.all.fa BmG -m primate7maml4.orthologous.ChA -t Bos_mutus.BosGru_v2.0.pep.all.modifiedtitle.fa`
 3.

2023新工作-使用新格式同源文件

/opt/pubdata/kakawork24primate/**primate7maml4.orthologous.ChA.BmG.Oar.NLeu.GSM.fornewformat.step1tout**

- 对误操作文件修正ID后得到 (primate7maml4.orthologous ->) primate7maml4.orthologous.ChA.BmG.Oar.NLeu.GSM
- sed 's/transcript\://g'
primate7maml4.orthologous.ChA.BmG.Oar.NLeu.GSM >
primate7maml4.orthologous.ChA.BmG.Oar.NLeu.GSM.fornew
format
- python
~/BDRobot/src/bioinfodevelop/analysisAppEntry/toolkit/one2
oneOrtholog.py -i
primate7maml4.orthologous.ChA.BmG.Oar.NLeu.GSM.fornew
format -s 1t -t TFhy -t GSM -b HMol -b lcJW -b MSP -b NLeu -
b sCA (使用得到
**primate7maml4.orthologous.ChA.BmG.Oar.NLeu.GSM.fornew
format.step1tout**文件,vim修掉title末尾的~)

准备文件完成后，重新计算kaks

```
(/opt/pubdata/kakawork24primate) $ python  
/opt/life/src/bioinfodevelop/analysisAppEntry/CalculateKaKsusingPaml.py -c /opt/pubdata/kakawork24primate/kaks.config -m  
/opt/pubdata/24pep/prime7maml4.orthologous.ChA.BmG.Oar  
.NLeu.GSM.fornewformat.step1tout [-s matchedanyway] -p  
/opt/pubdata/kakawork24primate/proteinCDSpair.txt -l 10 [-t  
/tmp/]
```